

Enterprise AI Customer Intelligence Platform

Solution Architecture & Implementation Guide

Enterprise Analytics Platform

Microsoft Fabric

Data Engineering • Artificial Intelligence • Business Intelligence

OneLake • Lakehouse • Warehouse • PySpark • Azure AI Foundry • GPT-5 • Power BI

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Solution Architecture & Implementation Guide

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1 Executive Summary

Key Takeaway

The Enterprise AI Customer Intelligence Platform demonstrates how Microsoft Fabric can be used to build an end-to-end enterprise analytics solution that combines Data Engineering, Artificial Intelligence and Business Intelligence within a modern Medallion Architecture.

Business Challenge

Organizations generate customer data from multiple systems including sales, products and customer reviews. While these datasets provide valuable business insights, they are often fragmented and customer feedback remains unstructured, limiting its analytical value.

Solution

The platform centralizes enterprise data within Microsoft Fabric using a Medallion Architecture. Customer reviews are enriched with Azure AI Foundry (GPT-5), transformed into structured business intelligence and loaded into a Fabric Warehouse using a Galaxy Schema. Power BI delivers interactive dashboards through a governed Semantic Model.

Technology Stack

Component	Technology
Data Integration	Fabric Data Factory
Storage	OneLake Lakehouse
Processing	PySpark
AI	Azure AI Foundry (GPT-5)
Warehouse	Fabric Warehouse
Analytics	Power BI Semantic Model
Reporting	Power BI
Deployment	Fabric Deployment Pipelines

Business Value

- Unified Customer 360 analytics
- AI-powered customer intelligence
- Enterprise-grade reporting
- Governed semantic model

- Scalable Microsoft Fabric architecture

2 Solution Architecture

Key Takeaway

The solution follows a layered enterprise architecture that separates data ingestion, engineering, AI enrichment, analytical storage and business reporting into independent services.

The platform integrates Microsoft Fabric services into a single analytical workflow. Data is ingested using Fabric Data Factory, stored within OneLake, transformed through the Medallion Architecture, enriched using Azure AI Foundry and loaded into a Fabric Warehouse before being consumed through a Power BI Semantic Model.

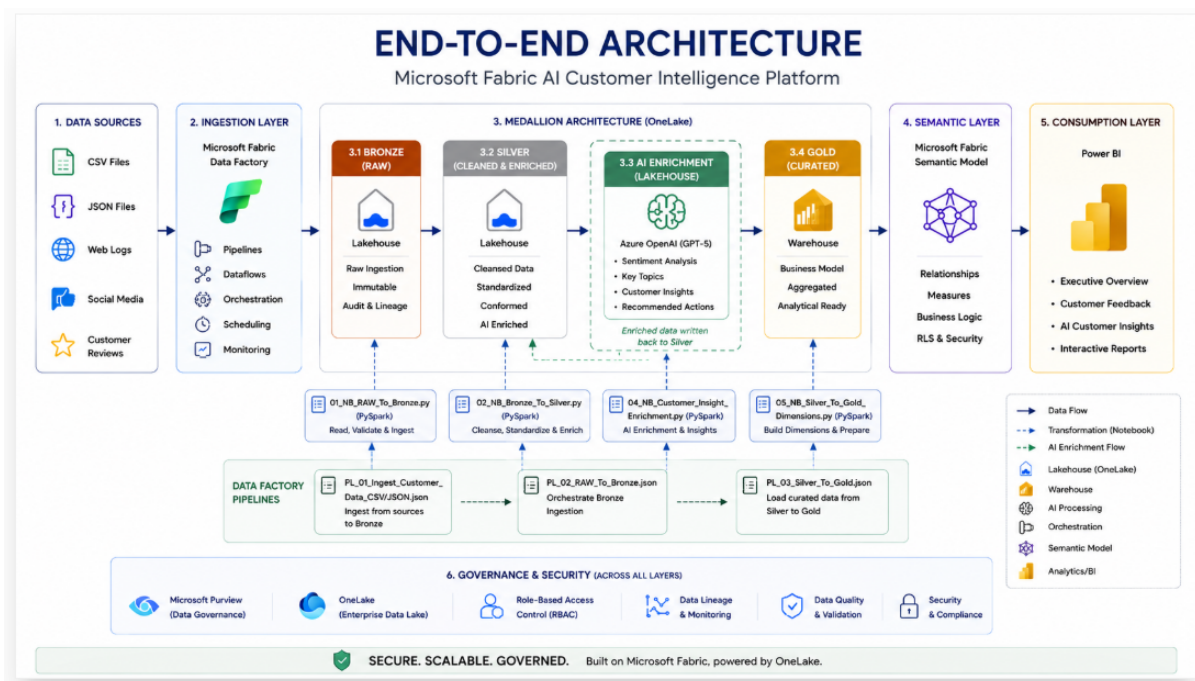


Figure 2.1. End-to-End Enterprise Architecture

Architecture Components

Component	Responsibility
Data Factory	Data ingestion
OneLake	Centralized storage
Lakehouse	Bronze and Silver layers
Azure AI Foundry	AI enrichment
Warehouse	Gold analytical model
Power BI	Business reporting

3 Medallion Architecture

Key Takeaway

The Medallion Architecture progressively improves data quality by separating raw ingestion, business transformation, AI enrichment and analytical modelling into dedicated processing layers.

The platform adopts Microsoft’s recommended Medallion Architecture to transform raw operational data into trusted analytical datasets.

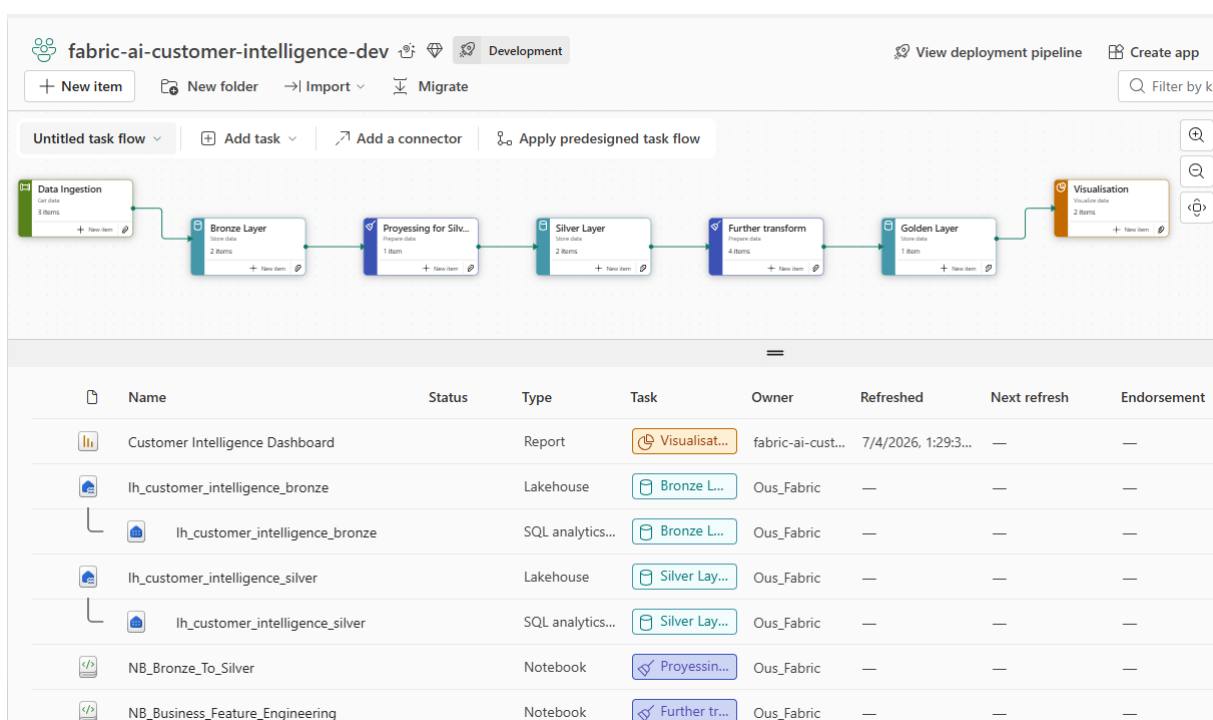


Figure 3.1. Microsoft Fabric Medallion Architecture

Processing Layers

Layer	Purpose
Bronze	Raw enterprise data
Silver	Cleansing and feature engineering
AI	Customer review enrichment
Gold	Business-ready analytical model

The separation of responsibilities simplifies governance, improves data quality and enables reusable analytical datasets.

4 AI Enrichment

Key Takeaway

Artificial Intelligence is integrated directly into the Silver layer, transforming customer reviews into structured business intelligence before loading the analytical warehouse.

Customer reviews are enriched using Azure AI Foundry (GPT-5). Rather than executing AI during reporting, enrichment occurs once during data engineering, ensuring consistent and reusable business insights.

Generated Attributes

- Sentiment
- Category
- Business Priority
- Summary
- Keywords
- Recommended Action

Business Benefits

- Consistent AI outputs
- Lower processing cost
- Faster dashboards
- Reusable enterprise intelligence
- Centralized AI governance

Example Output

Negative • Delivery • High Priority

Recommendation: Escalate to Logistics Team

5 Data Warehouse & Semantic Model

Key Takeaway

A Galaxy Schema combined with a Power BI Semantic Model enables multiple business domains to share common dimensions while maintaining independent fact tables.

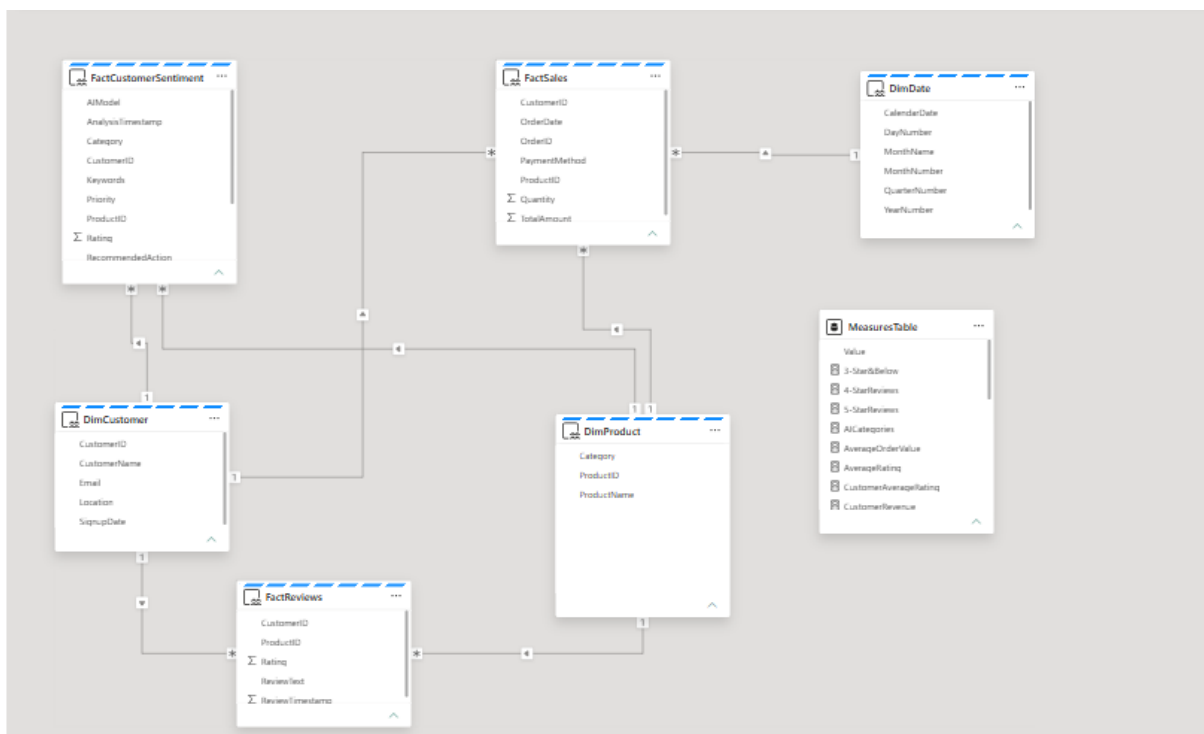


Figure 5.1. Enterprise Galaxy Schema

Analytical Objects

Object	Purpose
DimCustomer	Customer analytics
DimProduct	Product analytics
DimDate	Time intelligence
FactSales	Sales metrics
FactReviews	Review metrics
FactCustomerSentiment	AI insights
Measures Table	Centralized KPIs

6 Business Intelligence

Key Takeaway

Interactive Power BI dashboards transform curated analytical datasets into actionable business intelligence, enabling executives and operational teams to monitor performance, understand customer behaviour and identify AI-driven insights through a unified semantic model.

The Enterprise AI Customer Intelligence Platform delivers three interactive Power BI dashboards, each designed to answer specific business questions and support data-driven decision making.

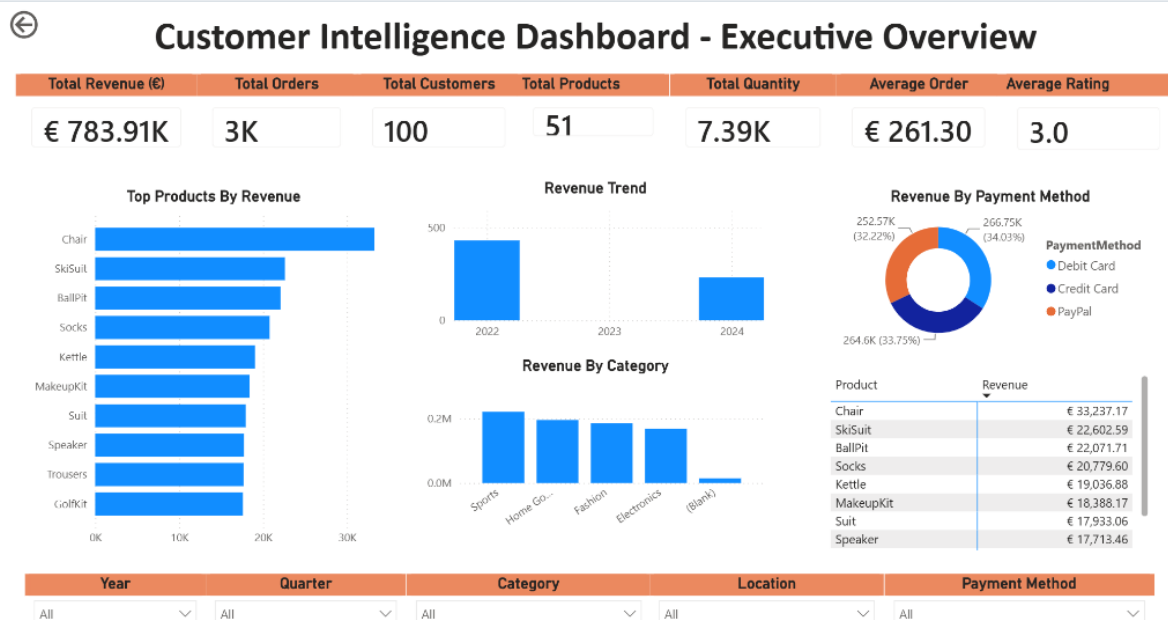


Figure 6.1. Executive Overview Dashboard

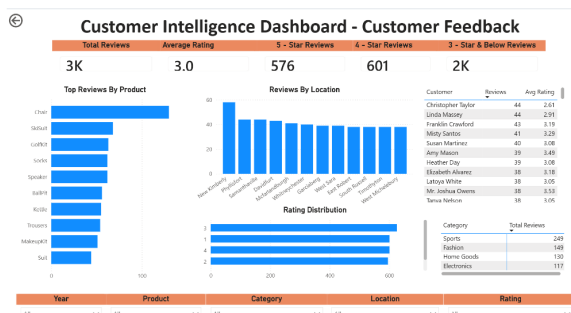


Figure 6.2. Customer Feedback Dashboard

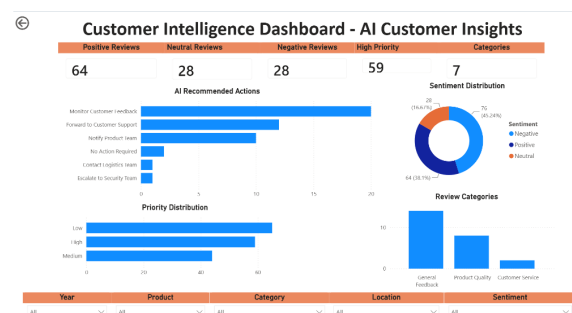


Figure 6.3. AI Customer Insights Dashboard

Business Questions Answered

Dashboard	Business Questions Answered
Executive Overview	Revenue performance, sales trends, customer satisfaction, top-performing products and executive KPIs.
Customer Feedback	Customer ratings, review distribution, product feedback, review trends and customer engagement.
AI Customer Insights	Customer sentiment, complaint categories, business priority, recommended actions and AI-generated insights.

7 Deployment Strategy

Key Takeaway

Microsoft Fabric Deployment Pipelines provide controlled promotion of analytical assets across Development, Test and Production environments.

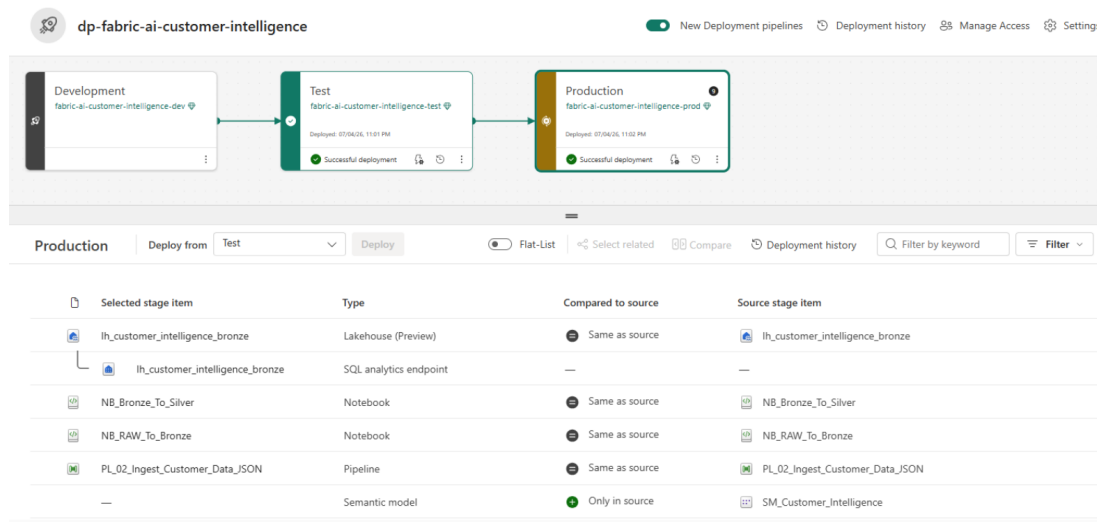


Figure 7.1. Deployment Pipeline

Current Implementation

Implemented	Planned
GitHub Repository	Fabric Git Integration
Azure DevOps Repository	Automated CI/CD
Deployment Pipelines	GitHub Actions

8 Conclusion

Key Takeaway

The Enterprise AI Customer Intelligence Platform demonstrates how Microsoft Fabric can be used to build scalable, AI-powered analytical solutions using modern enterprise architecture principles.

Key Achievements

- End-to-end Microsoft Fabric implementation
- AI-powered customer intelligence
- Enterprise Galaxy Schema
- Power BI Semantic Model
- Executive dashboards
- Deployment Pipelines

Future Enhancements

- Streaming ingestion
- Incremental refresh
- Fabric Git Integration
- Automated CI/CD
- RAG-powered customer support

Thank You

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